

Code Documentation: NPA matrix-valued polynomials / BFF

Key-Rate Estimation

for One-Sided Device-Independent QKD

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1 Method

This code lower-bounds the asymptotic key rate of a one-sided device-independent QKD protocol [1] by combining two ingredients. First, the conditional von Neumann entropy is bounded below using the BFF technique [2] via a Gauss–Radau quadrature sum over an auxiliary bounded operator $Z_{a,i}$ on Eve’s system for each outcome a and node i :

$$H(A|X=x_0, Q_E) \geq c_m + \sum_{i=1}^{m-1} \frac{w_i}{t_i \ln 2} \sum_a \inf_{Z_{a,i}} \text{Tr}[\rho(M_{a|x_0} \otimes (Z_{a,i} + Z_{a,i}^* + (1-t_i)Z_{a,i}^*Z_{a,i}) + t_i 1 \otimes Z_{a,i}Z_{a,i}^*)],$$

where (t_i, w_i) are the Gauss–Radau nodes/weights on $[0, 1]$ with $t_m = 1$ (`generate_quadrature`), and M sets the number of nodes used. Second, the moment matrix is built directly at the level of the untrusted party’s operator words, with each entry an $m \times m$ block $y_w \in \mathcal{B}(\mathbb{C}^m)$, $m = \text{DIMENSION}$, acting on the trusted party’s Hilbert space – the NPA [3] hierarchy with matrix-valued polynomials [1].

Parts of the BFF entropy-bound implementation are adapted from the code of P. Brown [4], in particular the Gauss–Radau quadrature evaluation of the entropy bound, the associated quadrature coefficients, and the construction of the BFF SDP objective at each quadrature node.

2 Bell Scenario Description

`SCENARIO` is a list of lists. The first element is a list containing the number of outputs for each input of the untrusted party (Alice or Bob, whichever is not `TRUSTED_PARTY`). The second element is a single-entry list containing the total number of Z and Z^* operators used in the BFF construction: these are the bounded operators acting on Eve’s subsystem.

Example: `SCENARIO = [[2, 2], [4]]` describes the untrusted party with two binary-outcome inputs (CHSH-type) and 4 total Z/Z^* operators on Eve’s subsystem, corresponding to the 2 outcomes of the key-generating measurement (Z_0, Z_1, Z_0^*, Z_1^*) . This version assumes the trusted and untrusted parties share the same number of inputs and outputs (as in CHSH/BB84); differing counts require minor changes in the code.

3 constraint_task_list Encoding

Both the imposed constraint (e.g. CHSH score) and the BFF objective at each quadrature node are described by lists of terms

[coefficient, matrix, word],

where *matrix* is the trusted party’s `DIMENSION × DIMENSION` POVM element for that term (or the identity, if the term has no trusted-party outcome), and *word* is the corresponding untrusted-party operator word. Each term contributes `coefficient · Tr[matrix ·  $\Gamma_{\text{word}}$ ]` to the SDP expression (`generate_objective_function_matrix`). Two constructions are included: `chsh_game_form_matrix` (CHSH winning probability for each input) and `bb84_matrix` (BB84 matching probability); other Bell scenarios and constraints can be supplied in the same format with minimal modifications.

4 Usage

Set `SCENARIO`, `LEVEL` (NPA level), `TRUSTED_PARTY`, `TRUSTED_PROJECTORS`, `M` (quadrature nodes), and the score sweep (`WMIN/WMAX/NPOINTS`); choose or supply `constraint_task_list`. Running the script solves the NPA matrix-valued polynomials/BFF SDP sweep and writes results to `plot_data.txt`. The estimated key rate can also be plotted against the calculated analytic fully-device-independent CHSH key rate [5], which coincides with the one-sided result only when the trusted party is also the key-generating party.

References

- [1] A. Bluhm, G. Koßmann, M. Sandfuchs, R. Schwonnek, G. Viola, R. Wolf, *Computing key rates for one-sided device-independent quantum key distribution* (2026).
- [2] P. Brown, *Device-independent lower bounds on the conditional von Neumann entropy – User guide* (2021).
- [3] M. Navascués, S. Pironio, A. Acín, *Bounding the Set of Quantum Correlations*, Phys. Rev. Lett. **98**, 010401 (2007).
- [4] P. Brown, *DI-rates*, GitHub repository (2021). <https://github.com/peterjbrown519/DI-rates>.
- [5] S. Pironio, A. Acín, N. Brunner, N. Gisin, S. Massar, V. Scarani, *Device-independent quantum key distribution secure against collective attacks*, New J. Phys. **11**, 045021 (2009).